Preparation And Characterization Of Lead Oxide Thin Films By Chemical Bath Deposition Technique

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***Abstract*— In this present work we grew plumbous oxide (lead oxide) thin films on suitably cleaned glass substrates by employing low cost chemical route such as chemical bath deposition. The lead oxide thin films were prepared using two different precursor solutions such as lead nitrate and lead acetate. The effects of deposition behaviour towards the properties of the thin films were investigated. The structural, morphological, and optical properties of thin films had been investigated by using XRD, SEM, and UV-Vis spectroscopy respectively. The XRD pattern of prepared lead oxide thin films was found to be in tetragonal structure with the preferred orientation along the plane (012). The SEM image reveals that the films were got the wings and irregular shaped structure. The optical band gap of the prepared thin films is 3.8 eV. The vibrational modes of the lead oxide thin films were also studied by micro Raman spectroscopic analysis and are well coincide with the earlier reporter.**

Keywords*— *Plumbous oxide; chemical bath deposition; band gap

I. INTRODUCTION

Transition metal oxide materials have been interesting of researchers, because of unique electrical and optical properties. In the recent years, increased attentions have been focused on the synthesis of lead oxide thin films. Among these, Lead oxide is an attractive semiconducting material have wide band gap, low electrical conductivity, interesting semiconducting and photo conducting properties, excellent chemical and thermal stability [1] which is widely used as transparent electrodes, solar cells, electro luminescent devices [2] microelectronic and opto electronic laser technology and imaging device applications [3] due to its unique electrical and optical properties [4]. Lead oxide thin films have been prepared by various techniques such as metal organic chemical vapor deposition [5], photo chemical metal organic deposition [6], atomic layer deposition [7], electro deposition [8-9], thermal evaporation [10]. Among the chemical methods chemical bath technique is very simple and inexpensive. Considering these advantages, in this work lead oxide thin films were synthesized by chemical bath techniques and the structural, morphological, optical and electrical properties of the films were studied and the results are presented here.

II. EXPERIMENTAL DETAILS

Lead oxide thin films were deposited on glass substrates using chemical bath deposition technique by lead nitrate and lead acetate precursors. Prior to deposition, the glass substrates were cleaned with chromic acid, ethanol, detergent, deionised water and acetone. The precursor solutions were prepared using molar ratio such as (0.05 : 0.1 :0.1) ($Pb(NO_3)_2$: NaOH : EDTA) and (0.05 : 0.1 : 0.1) (lead acetate : NaOH : EDTA). These materials dissolved in 25 ml of deionized water for preparation of PbO – thin films. The above solution will be deposited onto the glass substrate at different bath temperature (Room, 50 °C, 70 °C). The solution is stirred continuously for an hour with the help of magnetic stirrer. In which the aqua solution of ammonia was added drop by drop to reduces lead ions to lead hydroxides. The cleaned substrate was placed in the substrate holder at different desired time like (30, 60, 90, mins). Further the obtained thin films are annealed at 450 °C and are characterized by X-ray Diffraction, Micro Raman Analysis, Photoluminescence spectra, UV-Vis Spectra, Scanning Electron Microscope.

III. RESULTS AND DISCUSSION

Fig. 1(A) shows the X-ray diffraction pattern of PbO thin film synthesized by chemical bath deposi-

tion using lead nitrate $Pb(NO_3)_2$ as raw material at room temperature. All the observed diffraction peaks are indexed and good accordance with JCPDF card no: 76-1791. The peaks observed at 10.62, 12.78 and 32.45 ° are correspond to (012), (100) and (110) planes, respectively. All the prepared thin films are being polycrystalline with monoclinic structure with preferred orientation along (012) plane. The XRD patterns of the samples A1, A2 and A3 prepared at different bath temperature and annealed at 450 °C are shown in Fig. 1(A). The prominent peak observed at 31.65 corresponds to (110) plane, which is good agreement with JCPDF card no: A19-0680. The PbO thin films are polycrystalline with tetragonal structure.

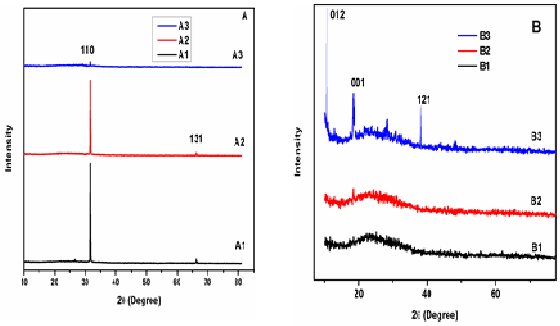

Fig..1 (A) X-ray Diffraction patterns of PbO thin films annealed at 450 °C with different bath temperature A1-RT, A2-50 °C, A3-70 °C). (B). Diffraction patterns of PbO thin films annealed at 450 °C with different deposition time. (B1-30 mins, B2-60 mins and B3-90 mins).

From this XRD results, one can conclude that, when $Pb(OH)_2$ thin film is annealed at 450 °C, the structural changes are occurred from monoclinic to tetragonal structure. Which is confirmed from absence of $Pb(OH)_2$ peaks and appearance of PbO peak at 2θ =32.45 °. Fig. 1(B) represents the XRD patterns of the $Pb(OH)_2$ and PbO thin film prepared using lead acetate as raw material at different deposition times. The diffraction peaks are listed and well concurrence with JCPDS card no: 76-1791 and previous report [11, 13]. When thin film is prepared at 30 and 60 minutes, respectively, a peak at 2θ 18.37 corresponds to (001) plane. It represents that the prepared thin film is polycrystalline with tetragonal with along the (001) orientation. As deposition time is increased up to 90 minutes, the additional peak appeared at 10.67 is attributed to $Pb(OH)_2$ which belongs to monoclinic structure. From this XRD results, one can conclude that as deposition time is increased from 60 to 90 min, the peak intensity is also increased with full width at half maximum of 0.24 and 0.19, which is also confirmed that crystallinity of the thin film is increased with deposition times. The crystalline sizes are increased as the deposition time increases. The amorphous structure obtained at the less deposited time, when deposition time increases the amorphous nature converted into polycrystalline nature. The average crystalline sizes, were calculated by using Debye Scherer's formula for the films prepared at different precursor solution, $D=0.9\lambda/\beta\cos\theta$. Where,λ - wavelength of X-ray (1.5407Å), β- full width half maximum (FWHM), θ- Diffracting angle. The crystalline size is calculated for A1, A2 and A3 are 140, 97.59, 97.60 and 97.60 respectively. The dislocation density and microstrain of PbO thin films as calculated from the following equation and are given in Table 1(a,b). $\delta = 1/D^2$ (lines/m2), $\varepsilon = \beta \cos\theta /4$ (lines/m^2). Micro strain and dislocation density is not changed by the changing the bath temperature which remains 0.03 and 1.04 respectively, for asprepared sample 50 and 0.02. Similar properties observed while changing the time for PbO synthesis from lead acetate (Fig. 1B). So that in the forth coming analysis we are giving less priority to interpret the lead nitrate precursor growth thin films.

Table 1 (a). Calculated parameter values for the PbO thin films annealed at 450 °C and different bath temperature. (S1-As-prepared at RT, A1-RT with annealed, A3-50 °C, A5-70 °C).

Sample	Crystalline Size (nm)	Dislocation Density $\times 10^{14}$ (lines^{-2}m^{-4})	Micro Strain $\times 10^{-3}$ (lines^{-2}m^{-4})
A1	97.59	1.04	0.03
A2	97.60	1.04	0.03
A3	97.60	1.04	0.03

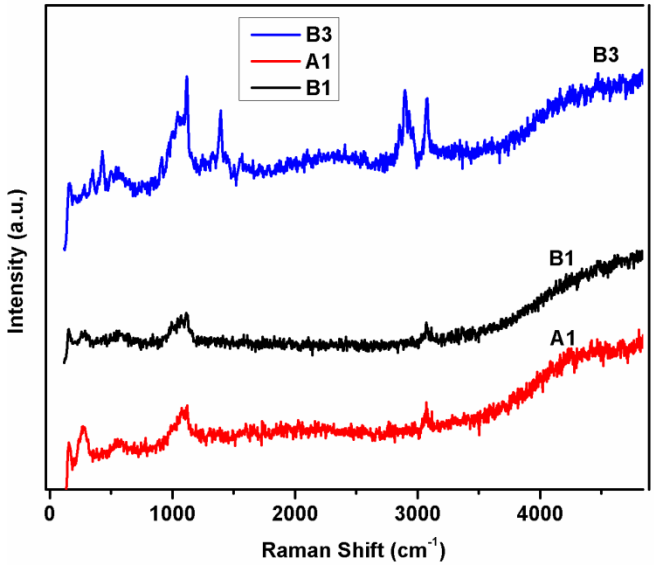

Fig. 2 Raman spectra of PbO thin films with different precursor solution annealed at 450 °C.

The characteristic vibrational micro Raman modes for lead oxide thin films are depict in Fig. 2 for a wave vector region 400 – 5000 cm^{-1}. The presence of three bands of micro Raman vibrational bands of carbonates are shown at 1050 cm^{-1}, 1053 cm^{-1}, 1056 cm-

[1]. Meanwhile a stretching band of the hydroxyl group noticed at 3139 cm^{-1}, 3162 cm^{-1} [12]. The band at 660 cm^{-1} attributed to stretching vibration of the PbO. There is no appreciable change of the PbO thin films [23]. The graphical diagram is concluded that the deposited time at 1.30 mins sample have good and sharp peaks. At the same time the lead nitrate solution samples have two bands at 1050and3139 cm^{-1}, which are vibrational band of carbonates and the band at 3139 cm^{-1} is stretching band of the hydroxyl group [12]. The peak position of PbO thin film before loading 282 cm^{-1} [22]. Comparing these samples of lead oxide thin films prepared by using the precursor solution of lead acetate samples get high intensity peaks. The annealing process caused no changes in the lead oxide phase is studied in this present study. The lead acetate solution of the sample B3 caused the dominant peaks at lead oxide [24]. The peak positions of the Raman spectra were shifted to the lower wave number [22].

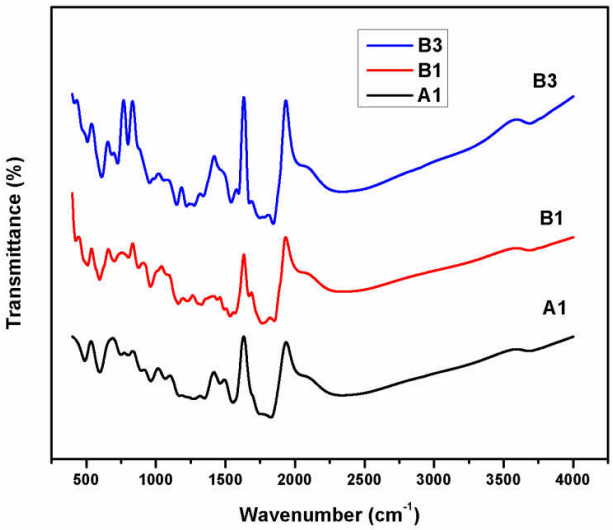

Fig. 3 FTIR spectra of PbO thin films with different precursor solution annealed at 450 °C. The FTIR spectra of PbO thin films prepared at different precursor materials were shown.

FTIR spectra of PbO thin film is in showed Fig.3 a broad and weak intensive band around at 3865 cm^{-1} which corresponds to the vibration mode of OH stretching group due to the small amount of water absorbed on the PbO thin film surface. A band at 2344 cm^{-1} and 1595 cm^{-1} is assigned due to C=C stretching vibration of EDTA for asprepared sample (A1). A sharp and intensive peak observed at 510 cm^{-1} is attributed to the antisymmetric bending vibrations of Pb-O-Pb chain present the film (A1). The weak peak observed at 415 and 424 cm^{-1} are related to the bending vibration of PbO [27, 28].

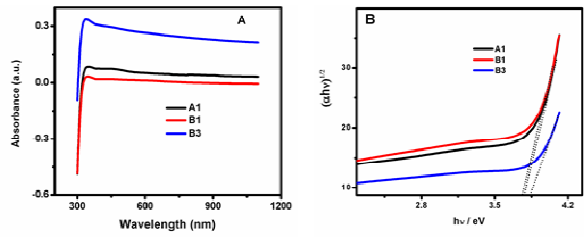

Fig. 4 (A) Optical absorption spectra of PbO thin films with different precursor solution annealed at 450 °C and (B) is Tauc plot to estimate bandgap energy

The optical absorption spectra of the PbO thin films deposited onto glass substrate were studied in the scanning range of wavelengths 300-1000 nm is shown in Fig. 4. The calculated refractive index of the lead oxide thin film is depicting at 1.5. The thickness of the sample A1 is 27.96 µm. The thickness of the sample B1 and B3 are the 25.61 µm and 55.11 µm. Comparing these two precursors and their basic components lead acetate is better than the film deposit by lead oxide thin films. This can be interpreted as opacity of the film. This is the indirect band gap transitions. These absorption spectra, are the most and perhaps the simplest method for probing the band structure of semiconductors, are employed in the determination of the energy gap Eg. The band gap energy was not changed by the changing precursor and experimental condition. The obtained PbO energy band gap is 3.8 eV.

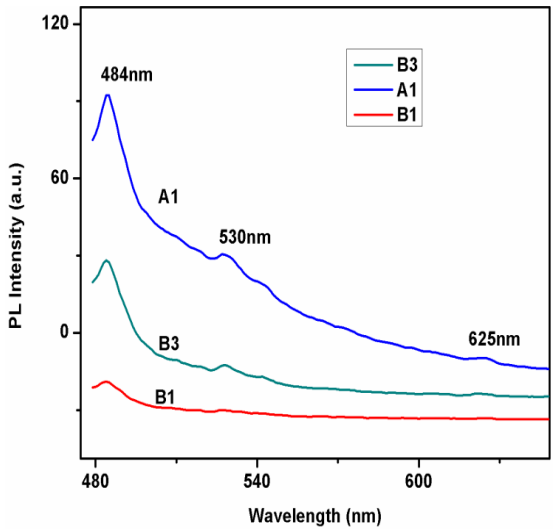

Fig. 5. PL emission spectra of PbO thin films prepared with different precursor solution annealed at 450 °C.

A typical up conversion emission spectra of lead oxide thin films deposited at two different precursor is shown in Fig.5 and it is observed that there are three distinctive emission peak centred at 530 nm, 484.7 nm, and 625 nm its due to possibility of the blue and green emission respectively. The weak green emission

band was carried at 530 nm where as a weak red emission band is observed at 605 nm. In the reference the strong green emission peak at 560 nm [25]. In all these luminous spectra there are two emission peaks appeared at 457.9 nm and 530 nm which are being in the deep level visible region [20]. The weak emissive with less intensive peak of PbO thin films are observed at 530 nm. The sharp intensive emission peak associated with high intensity is appeared at around 457.9 nm due to exciton present in a donor energy level. This process leads to the recombination luminescence and stoichiometry condition of prepared PbO thin films [25, 26]. The higher luminescence properties are observed at when the film prepared by $Pb(NO_3)_2$ precursor solution.

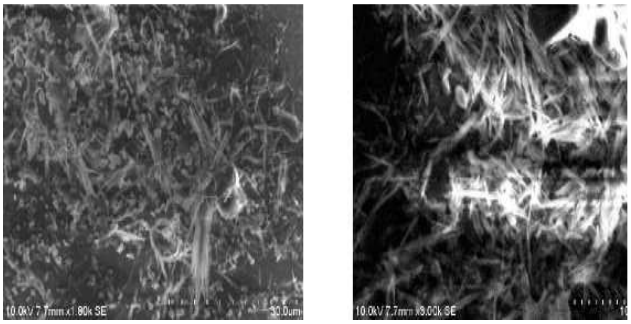

Fig. 6. (a) SEM images of lead oxide thin films prepared using lead nitrate with different magnification.

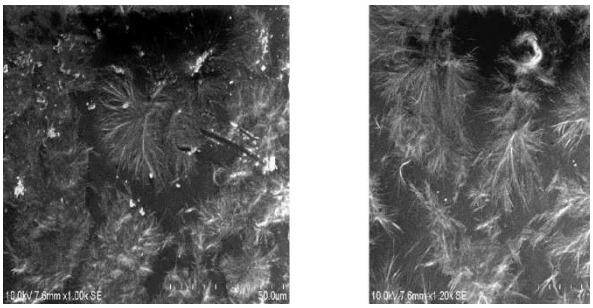

Fig.6. (b) SEM images for lead oxide thin films were prepared using lead acetate with different magnification.

Fig. 6 shows scanning electron micrographs of lead oxide thin films deposited by chemical bath deposition technique at different precursor. The SEM image for lead oxide thin films prepared by using lead nitrate as shown in fig.6a PbO has an irregular shaped structure. However, some needle shaped structure is observed between grains in the SEM micrograph. The SEM image for lead oxide thin films prepared by using lead acetate as a precursor material as shown in fig. 6b lead oxide had a softly in wings shaped structure. Some of the agglomeration particles are visualized in the structure [17, 18]. By changing of precursor, morphologies of the lead oxide thin films are changed [14 -16].

IV. Conclusion

The lead oxide thin films were prepared at different precursor materials using chemical bath deposition technique. The structural properties of the lead oxide thin films were studied by using X-Ray diffraction pattern. From the X-Ray diffraction analysis of the PbO thin films were found to the tetragonal structure both precursors. The emission property of the film was studied by photoluminescence spectroscopy slights changes are observed by the varying experiment conditions. The functional groups of the lead oxide thin films were analyzed by FTIR spectroscopy. Vibrational bands of carbonates and stretching band of hydroxyl of the lead oxide thin films were identified by Raman spectroscopy. The optical properties of the PbO thin films were studied from UV-Vis transmission spectroscopy; the calculated band gap is 3.7-3.8 eV. The surface morphological of the lead oxide thin films was studied by Scanning Electron Microscopy.

Acknowledgment

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